Java Practicals - Short Code Sheet

Short, easy-to-write versions of all 10 practicals. NetBeans 8.2 + GlassFish 4.1.1 (javax.* only). Class name = file name. HTML form action = servlet name. EJB practicals (2, 4, 8, 10): Enterprise Application with EJB module + Web module, and add the EJB module as a dependency of the web module.

Common imports for every servlet:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.*;
import javax.ejb.*;          // only for EJB practicals
```

1. Simple Calculator (Servlet)

index.html

```
<form action="CalcServlet" method="post">
No 1: <input name="a"> No 2: <input name="b">
<select name="op"><option>+</option><option>-</option><option>*</option><option>/</option></select>
<input type="submit">
</form>
```

CalcServlet.java

```
@WebServlet("/CalcServlet")
public class CalcServlet extends HttpServlet {
  protected void doPost(HttpServletRequest req, HttpServletResponse res)
      throws ServletException, IOException {
    PrintWriter out = res.getWriter();
    double a = Double.parseDouble(req.getParameter("a"));
    double b = Double.parseDouble(req.getParameter("b"));
    String op = req.getParameter("op");
    double r = 0;
    if (op.equals("+")) r = a + b;
    else if (op.equals("-")) r = a - b;
    else if (op.equals("*")) r = a * b;
    else r = a / b;
    out.println("Result: " + r);
  }
}
```

2. Currency Converter (EJB)

ConverterBean.java (EJB module)

```
@Stateless
public class ConverterBean {
  public double toInr(double usd) { return usd * 83; }
}
```

index.html (Web module)

```
<form action="CurrServlet" method="post">
USD: <input name="usd"> <input type="submit" value="Convert">
</form>
```

CurrServlet.java (Web module)

```
@WebServlet("/CurrServlet")
public class CurrServlet extends HttpServlet {
  @EJB ConverterBean bean;
  protected void doPost(HttpServletRequest req, HttpServletResponse res)
      throws ServletException, IOException {
    double usd = Double.parseDouble(req.getParameter("usd"));
    res.getWriter().println("INR: " + bean.toInr(usd));
  }
}
```

Rate 83 is an example. Use the rate your teacher gives. Import the bean's package if it is not the default package.

3. Password Validation (RequestDispatcher)

index.html

```
<form action="PassServlet" method="post">
Password: <input type="password" name="pw"> <input type="submit">
</form>
```

PassServlet.java

```
@WebServlet("/PassServlet")
public class PassServlet extends HttpServlet {
  protected void doPost(HttpServletRequest req, HttpServletResponse res)
      throws ServletException, IOException {
    if ("Admin".equals(req.getParameter("pw")))
      req.getRequestDispatcher("WelcomeServlet").forward(req, res);
    else {
      res.getWriter().println("Wrong password. <a href='index.html'>Try again</a>");
    }
  }
}
```

WelcomeServlet.java

```
@WebServlet("/WelcomeServlet")
public class WelcomeServlet extends HttpServlet {
  protected void doPost(HttpServletRequest req, HttpServletResponse res)
      throws ServletException, IOException {
    res.getWriter().println("Welcome! Password accepted.");
  }
}
```

forward() keeps the POST method, so WelcomeServlet must also have doPost.

4. Marks Entry (EJB + Database)

StudentMark.java (EJB module, JPA entity)

```
@javax.persistence.Entity
public class StudentMark implements java.io.Serializable {
  @javax.persistence.Id @javax.persistence.GeneratedValue
  public Long id;
  public String name;
  public int marks;
}
```

MarksBean.java (EJB module)

```
@Stateless
public class MarksBean {
  @javax.persistence.PersistenceContext(unitName = "MarksPU")
  javax.persistence.EntityManager em;

  public void save(String name, int marks) {
    StudentMark s = new StudentMark();
    s.name = name; s.marks = marks;
    em.persist(s);
  }
}
```

persistence.xml (EJB module, META-INF)

```
<persistence version="2.1" xmlns="http://xmlns.jcp.org/xml/ns/persistence">
 <persistence-unit name="MarksPU" transaction-type="JTA">
  <jta-data-source>jdbc/sample</jta-data-source>
  <properties>
   <property name="eclipselink.ddl-generation" value="create-tables"/>
  </properties>
 </persistence-unit>
</persistence>
```

index.html

```
<form action="MarksServlet" method="post">
Name: <input name="name"> Marks: <input name="marks">
<input type="submit" value="Save">
</form>
```

MarksServlet.java (Web module)

```
@WebServlet("/MarksServlet")
public class MarksServlet extends HttpServlet {
  @EJB MarksBean bean;
  protected void doPost(HttpServletRequest req, HttpServletResponse res)
      throws ServletException, IOException {
    bean.save(req.getParameter("name"), Integer.parseInt(req.getParameter("marks")));
    res.getWriter().println("Marks saved successfully.");
  }
}
```

jdbc/sample is GlassFish's default sample database resource. Use your lab's JDBC resource name if it is different.

5. Calculator using HTML and Servlet

Same as Practical 1. Use the same index.html and CalcServlet.java.

6. JSP - Insert into emp Table

SQL

```
CREATE TABLE emp (eno INT PRIMARY KEY, ename VARCHAR(100), salary DECIMAL(10,2));
```

employee.jsp

```
<%@ page import="java.sql.*" %>
<form method="post">
Eno: <input name="eno"> Name: <input name="ename"> Salary: <input name="salary">
<input type="submit" value="Insert">
</form>
<%
if (request.getParameter("eno") != null) {
  try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/practicaldb", "root", "YOUR_PASSWORD");
    PreparedStatement ps = con.prepareStatement("INSERT INTO emp VALUES (?,?,?)");
    ps.setInt(1, Integer.parseInt(request.getParameter("eno")));
    ps.setString(2, request.getParameter("ename"));
    ps.setDouble(3, Double.parseDouble(request.getParameter("salary")));
    ps.executeUpdate();
    out.println("Record inserted.");
    con.close();
  } catch (Exception e) { out.println(e); }
}
%>
```

Add the MySQL Connector/J jar to the project Libraries. Older connector versions use com.mysql.jdbc.Driver.

7. Simple Validations (Servlet)

index.html

```
<form action="ValidServlet" method="post">
Name: <input name="name"> Email: <input name="email"> Age: <input name="age">
<input type="submit" value="Validate">
</form>
```

ValidServlet.java

```
@WebServlet("/ValidServlet")
public class ValidServlet extends HttpServlet {
  protected void doPost(HttpServletRequest req, HttpServletResponse res)
      throws ServletException, IOException {
    PrintWriter out = res.getWriter();
    String name = req.getParameter("name");
    String email = req.getParameter("email");
    try {
      int age = Integer.parseInt(req.getParameter("age"));
      if (name.trim().isEmpty()) out.println("Name is required");
      else if (!email.contains("@") || !email.contains(".")) out.println("Invalid email");
      else if (age < 1 || age > 120) out.println("Invalid age");
      else out.println("Validation successful. Welcome " + name);
    } catch (NumberFormatException e) { out.println("Age must be a number"); }
  }
}
```

8. Room Reservation (EJB)

RoomBean.java (EJB module)

```
@Stateless
public class RoomBean {
  public double cost(String type, int nights) {
    double rate = type.equals("Single") ? 1500 : type.equals("Double") ? 2500 : 4000;
    return rate * nights;
  }
}
```

index.html

```
<form action="RoomServlet" method="post">
Guest: <input name="guest">
Room: <select name="type"><option>Single</option><option>Double</option><option>Deluxe</option></select>
Nights: <input name="nights"> <input type="submit" value="Calculate">
</form>
```

RoomServlet.java (Web module)

```
@WebServlet("/RoomServlet")
public class RoomServlet extends HttpServlet {
  @EJB RoomBean bean;
  protected void doPost(HttpServletRequest req, HttpServletResponse res)
      throws ServletException, IOException {
    String type = req.getParameter("type");
    int n = Integer.parseInt(req.getParameter("nights"));
    PrintWriter out = res.getWriter();
    out.println("Guest: " + req.getParameter("guest") + "<br>Room: " + type);
    out.println("<br>Nights: " + n + "<br>Total: Rs. " + bean.cost(type, n));
  }
}
```

9. Online Exam Login (Servlet)

index.html

```
<form action="LoginServlet" method="post">
User: <input name="user"> Password: <input type="password" name="pw">
<input type="submit" value="Login">
</form>
```

LoginServlet.java

```
@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {
  protected void doPost(HttpServletRequest req, HttpServletResponse res)
      throws ServletException, IOException {
    PrintWriter out = res.getWriter();
    if ("student".equals(req.getParameter("user")) && "Exam123".equals(req.getParameter("pw")))
      out.println("Login successful. Welcome to the exam portal.");
    else
      out.println("Invalid username or password. <a href='index.html'>Try again</a>");
  }
}
```

Test login: student / Exam123.

10. Hotel Reservation (EJB, with Check-in Date)

HotelBean.java (EJB module)

```
@Stateless
public class HotelBean {
  public double total(String type, int nights) {
    double rate = type.equals("Standard") ? 2000 : type.equals("Deluxe") ? 3500 : 6000;
    return rate * nights;
  }
}
```

index.html

```
<form action="HotelServlet" method="post">
Name: <input name="guest"> Email: <input name="email">
Room: <select name="type"><option>Standard</option><option>Deluxe</option><option>Suite</option></select>
Check-in: <input type="date" name="date"> Nights: <input name="nights">
<input type="submit" value="Reserve">
</form>
```

HotelServlet.java (Web module)

```
@WebServlet("/HotelServlet")
public class HotelServlet extends HttpServlet {
  @EJB HotelBean bean;
  protected void doPost(HttpServletRequest req, HttpServletResponse res)
      throws ServletException, IOException {
    String type = req.getParameter("type");
    int n = Integer.parseInt(req.getParameter("nights"));
    PrintWriter out = res.getWriter();
    out.println("<h3>Reservation Summary</h3>");
    out.println("Guest: " + req.getParameter("guest") + "<br>Email: " + req.getParameter("email"));
    out.println("<br>Room: " + type + "<br>Check-in: " + req.getParameter("date"));
    out.println("<br>Nights: " + n + "<br>Total: Rs. " + bean.total(type, n));
  }
}
```

Result Lines (write at end of each practical)

1. Thus, the calculator application using Servlet was created and executed successfully.

2. Thus, the currency converter using EJB was implemented successfully.
3. Thus, the requestDispatcher was used to validate the password and forward to WelcomeServlet.
4. Thus, the marks entry application saved student marks through an EJB and database.
5. Thus, the the HTML form called a Servlet to calculate and display the result.
6. Thus, the the JSP page inserted employee details into the emp table.
7. Thus, the server-side form validation was demonstrated using a Servlet.
8. Thus, the room reservation cost was calculated through an EJB.
9. Thus, the examination login Servlet displayed a success or failure message.
10. Thus, the hotel reservation application accepted guest and stay details and displayed an estimate.